Kinetic Energy

$$T = \sum_I \int^{r_{max}} d^3r \frac{3(6\pi^2\hbar^3)^{5/3}}{20\pi^2\hbar^3 m^I (g^I)^{2/3}} (n^I(r))^{5/3}$$

$$+ \sum_I \int^{r_{max}} d^3r \frac{3(6\pi^2\hbar^3)^{5/3}}{20\pi^2\hbar^3 \bar{m}^I (\bar{g}^I)^{2/3}} (\bar{n}^I(r))^{5/3}$$

We have,

$$n^I(r) = \left(\frac{2\times(4/3)\alpha_s}{a}\right)^{3/2} \left(\frac{f^I(r)}{r}\right)^{3/2} \frac{g^I}{6\pi^2}$$

$$\bar{n}^I(r) = \left(\frac{2\times(4/3)\alpha_s}{a}\right)^{3/2} \left(\frac{\bar{f}^I(r)}{r}\right)^{3/2} \frac{\bar{g}^I}{6\pi^2}$$

So that

$$T = \sum_I \int^{r_{max}} d^3r \frac{3(6\pi^2\hbar^3)^{5/3}}{20\pi^2\hbar^3 m^I (g^I)^{2/3}} \left\{\left(\frac{2\times(4/3)\alpha_s}{a}\right)^{\frac{3}{2}\times\frac{5}{3}} \left(\frac{f^I(r)}{r}\right)^{\frac{3}{2}\times\frac{5}{3}} \frac{(g^I)^{5/3}}{(6\pi^2)^{5/3}}\right.$$

$$\left. + \sum_I \int^{r_{max}} d^3r \frac{3(6\pi^2\hbar^3)^{5/3}}{20\pi^2\hbar^3 \bar{m}^I (\bar{g}^I)^{2/3}} \left\{\left(\frac{2\times(4/3)\alpha_s}{a}\right)^{\frac{3}{2}\times\frac{5}{3}} \left(\frac{\bar{f}^I(r)}{r}\right)^{\frac{3}{2}\times\frac{5}{3}} \frac{(\bar{g}^I)^{5/3}}{(6\pi^2)^{5/3}}\right\}\right.$$

$$= \sum_I \int^{r_{max}} d^3r \frac{3\hbar^2 g^I}{20\pi^2 m^I} \left(\frac{2\times\frac{4}{3}\alpha_s}{a}\right)^{\frac{5}{2}} \left(\frac{f^I(r)}{r}\right)^{\frac{5}{2}}$$

$$+ \sum_I \int^{r_{max}} d^3r \frac{3\hbar^2 \bar{g}^I}{20\pi^2 \bar{m}^I} \left(\frac{2\times\frac{4}{3}\alpha_s}{a}\right)^{\frac{5}{2}} \left(\frac{\bar{f}^I(r)}{r}\right)^{\frac{5}{2}}$$

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Let $T = T_1 + T_2$

So that,

$$T_1 = \sum_I \int_0^{r_{max}} d^3r \; \frac{3\hbar^2 g^I}{20\pi^2 m^I} \left(\frac{2\times\frac{4}{3}\alpha_s}{a}\right)^{\frac{5}{2}} \left(\frac{f^I(r)}{r}\right)^{\frac{5}{2}}$$

$$T_2 = \sum_I \int_0^{r_{max}} d^3r \; \frac{3\hbar^2 g^{-I}}{20\pi^2 m^I} \left(\frac{2\times\frac{4}{3}\alpha_s}{a}\right)^{\frac{5}{2}} \left(\frac{f^I(r)}{r}\right)^{\frac{5}{2}}$$

Taking T_1,

$$T_1 = \frac{3\hbar^2}{20\pi^2 m_1}\left(\frac{2\times\frac{4}{3}\alpha_s}{a}\right)^{\frac{5}{2}} \sum_I \int_0^{r_{max}} d^3r \, \alpha_I g^I \left(\frac{f^I(r)}{r}\right)^{\frac{5}{2}}$$

$$= \underbrace{\frac{3\hbar^2}{20\pi^2 m_1}\left(\frac{2\times\frac{4}{3}\alpha_s}{a}\right)^{\frac{5}{2}} 4\pi \sum_I}_{\text{Constant}} \int_0^{r_{max}} r^2 dr \, \alpha_I g^I \left(\frac{f^I(r)}{r}\right)^{\frac{5}{2}}$$

$$= \text{Constant} \cdot \sum_I \int_0^{r_{max}} dr \; r^2 \alpha_I g^I \left(\frac{f^I(r)}{r}\right)^{\frac{5}{2}}$$

Put $r = Rx$

$dr = R\,dx$

$$= \text{Constant} \cdot \sum_I \int_0^{x_{max}} R\,dx\, R^2 x^2 \alpha_I g^I \left(\frac{f^I(Rx)}{Rx}\right)^{5/2}$$

$$= \text{Constant} \cdot \sum_I \int_0^{x_{max}} dx \, R^{3-\frac{5}{2}} \alpha_I g^I \frac{\left(f^I(x)\right)^{5/2}}{x^{5/2-2}}$$

$$= \text{Constant} \cdot \sum_I \int_0^{x_{max}} dx \sqrt{R}\, \alpha_I g^I \frac{\left(f^I(x)\right)^{5/2}}{\sqrt{x}}$$

$$= (\underbrace{\text{Constant}}_{C}) \sum_I \int_0^{x_{max}} dx \, \alpha_I g^I \frac{\left(f^I(x)\right)^{5/2}}{\sqrt{x}}$$

This Constant term contains additional term $\sqrt{R}$

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$$T_I = C \sum_I \alpha_I \xi^I \int_0^{x_{max}} dx \frac{(f^I(x))^{5/2}}{\sqrt{x}} \quad (3)$$

$$\text{or, } \frac{T_I}{C} = \sum_I \alpha_I \xi^I \int_0^{x_{max}} dx \frac{(f^I(x))^{5/2}}{\sqrt{x}} \quad \ldots\ldots (A)$$

$$= \alpha_I \xi_I \int_0^{x_2} dx \frac{(f_2^I(x))^{5/2}}{\sqrt{x}} + \alpha_2 \xi_2 \int_0^{x_1} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}}$$

$$+ \alpha_1 \xi_1 \int_0^{x_1} dx \, f_1(x)$$

$$= \alpha_I \xi_I \int_0^{x_2} dx \frac{(f^I(x))^{5/2}}{\sqrt{x}} + \alpha_I \xi_I \int_0^{x_2} dx \frac{(f^I(x))^{5/2}}{\sqrt{x}} + \alpha_1 \xi_I \int_0^{x_1} dx \frac{(f^I(x))^{5/2}}{\sqrt{x}}$$

$$\frac{T_I}{C} = \sum_I \alpha_I \xi^I \int_0^{x_{max}} dx \frac{(f^I(x))^{5/2}}{\sqrt{x}}$$

$$= \alpha_2 \xi_2 \int_0^{x_2} dx \frac{(f_2^I(x))^{5/2}}{\sqrt{x}} + \alpha_1 \xi_1 \int_0^{x_1} dx \frac{(f_1(x))^{5/2}}{\sqrt{x}}$$

$$= \alpha_2 \xi_2 \int_0^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}} + \alpha_2 \xi_2 \int_{x_2}^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}}$$

$$+ \alpha_1 \xi_1 \int_0^{x_2} dx \frac{(f_1(x))^{5/2}}{\sqrt{x}} + \alpha_1 \xi_1 \int_{x_2}^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}}$$

$$+ \alpha_1 \xi_1 \int_{x_2}^{x_1} dx \frac{(f_1(x))^{5/2}}{\sqrt{x}}$$

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$$= \alpha_2 \xi_2 \beta_{22}^{5/2} \int_0^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}} + \alpha_2 \xi_2 \int_{x_2}^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}} \quad (4)$$

$$+ \alpha_1 \xi_1 \beta_{12}^{5/2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \alpha_1 \xi_1 \Gamma_{12}^{5/2} \int_{x_2}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$+ \alpha_1 \xi_1 \int_{x_2}^{x_1} dx \frac{(f_1(x))^{5/2}}{\sqrt{x}}$$

$$(\alpha_2 \xi_2 \beta_{22}^{5/2} + \alpha_1 \xi_1 \beta_{12}^{5/2}) \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$+ (\alpha_2 \xi_2 + \alpha_1 \xi_1 \Gamma_{12}^{5/2}) \int_{x_2}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \alpha_1 \xi_1 \int_{x_2}^{x_1} dx \frac{(f_1(x))^{5/2}}{\sqrt{x}}$$

From appendix,

$$\int_0^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}} = \frac{4}{7} \sqrt{x_2} (f_2(x))^{5/2} - \frac{5}{21\eta} N_2 \frac{df_2(x)}{dx}$$

$$\int_{x_2}^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}} = \frac{4}{7} \sqrt{x_2} (f_2(x))^{5/2} - \frac{4}{7} \sqrt{x_2} (\beta_{22})^{5/2} (f_2(x))^{\frac{5}{2}}$$

$$- \frac{5}{7 Q_2} \frac{df_2(x)}{dx} \frac{N_2 Q_2}{3\eta} + \frac{5}{7 Q_2} \beta_{22}^2 \frac{df_2(x)}{dx} \frac{N_2 Q_2}{3\eta}$$

$$\int_{x_2}^{x_1} dx (f_1(x))^{5/2} = \frac{4}{7} \sqrt{x_1} (f_1(x))^{5/2} - \frac{4}{7} \Gamma_{12}^{5/2} \sqrt{x_2} (f_2(x))^{5/2}$$

$$- \frac{5}{21\eta} \left[N_1 - N_2 \left(\Gamma_{12}^{3/2} - \frac{Q_2}{Q_1} \Gamma_{12} \right) \right] \frac{df_1(x)}{dx}$$

$$+ \frac{5}{21\eta} \Gamma_{12}^2 N_2 \frac{Q_2}{Q_1} \frac{df_2(x)}{dx}$$

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$$= (\alpha_2 \xi_2 \beta_{22}^{5/2} + \alpha_1 \xi_1 \beta_{12}^{5/2}) \left[\frac{4}{7} \sqrt{x_2} (f_2(x))^{5/2} - \frac{5 N_2}{2 \eta} \frac{df_2(x)}{dx} \right]$$

$$+ (\alpha_2 \xi_2 + \alpha_1 \xi_1 \gamma_{12}^{5/2}) \left[\frac{4}{7} \sqrt{x_2} (f_2(x))^{\frac{5}{2}} - \frac{4}{7} \sqrt{x_2} (\beta_{22})^{5/2} (f_2(x))^{\frac{5}{2}} \right.$$

$$\left. - \frac{5}{7 Q_2} \frac{df_2(x)}{dx} \frac{N_2 Q_2}{3\eta} + \frac{5}{7 Q_2} \beta_{22}^2 \frac{df_2(x)}{dx} \frac{N_2 Q_2}{3\eta} \right]$$

$$+ \alpha_1 \xi_1 \left[\frac{4}{7} \sqrt{x_1} (f_1(x))^{5/2} - \frac{4}{7} \gamma_{12}^{5/2} \sqrt{x_2} (f_2(x))^{5/2} \right.$$

$$\left. - \frac{5}{2 \eta} \left\{ N_1 - N_2 \left(\gamma_{12}^{3/2} - \frac{Q_2}{Q_1} \gamma_{12} \right) \right\} \frac{df_1(x)}{dx} \right.$$

$$\left. + \frac{5}{2 \eta} \gamma_{12}^2 N_2 \frac{Q_2}{Q_1} \frac{df_2(x)}{dx} \right] \quad \ldots\ldots (B)$$

Taking T_2/c, (Making use of eqn ... (A))

$$\frac{T_2}{c} = \alpha_2 \xi_2 \int_0^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}}, \quad \leftarrow \text{From appendix}$$

$$= \alpha_2 \xi_2 \left[\frac{4}{7} \sqrt{x_2} (f_2(x))^{5/2} - \frac{5 N_2}{2 \eta} \frac{df_2(x)}{dx} \right]$$

Now, adding $\frac{T_1}{c} + \frac{T_2}{c}$ to get the final expression of total kinetic energy. We will multiply the term 'c' at last for simplicity.

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Total kinetic energy

$$T = T_1 + T_2$$

$$= \left(\alpha_2 \xi_2 \beta_{22}^{5/2} + \alpha_1 \xi_1 \beta_{12}^{5/2} + \alpha_2 \xi_2\right)$$

$$\left[\frac{4}{7}\sqrt{x_2}\,(f_2(x))^{5/2} - \frac{5N_2}{2\eta}\frac{df_2(x)}{dx}\right]$$

$$+ \left(\alpha_2 \xi_2 + \alpha_1 \xi_1 r_{12}^{5/2}\right)\left[\frac{4}{7}\sqrt{x_2}\,(f_2(x))^{5/2}\right.$$

$$-\frac{4}{7}\sqrt{x_2}\,(\beta_{22})^{5/2}(f_2(x))^{5/2} - \frac{5}{7}\frac{df_2(x)}{dx}\frac{N_2}{3\eta}$$

$$\left. + \frac{5}{7Q_2}\beta_{22}^2\frac{df_2(x)}{dx}\frac{N_2 Q_2}{3\eta}\right] + \alpha_1 \xi_1 \left[\frac{4}{7}\sqrt{x_1}\right.$$

$$(f_1(x))^{5/2} - \frac{4}{7} r_{12}^{5/2}\sqrt{x_2}\,(f_2(x))^{5/2} - \frac{5}{2\eta}\left\{N_1\right.$$

$$\left. - N_2\left(r_{12}^{3/2} - \frac{Q_2}{Q_1} r_{12}\right)\right\}\frac{df_1(x)}{dx} + \frac{5}{2\eta} r_{12}^2$$

$$\left. N_2 \frac{Q_2}{Q_1}\frac{df_2(x)}{dx}\right].$$

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Appendix

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$$\int_0^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}}, \quad \int_{x_2}^{x_2} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}}, \quad \int_{x_2}^{x_1} dx \frac{(f_2(x))^{5/2}}{\sqrt{x}}$$

$$\int \frac{dx (f(x))^{5/2}}{\sqrt{x}} = \frac{4}{7}\sqrt{x}(f(x))^{5/2} - \frac{5}{7Q} f'[xf' - f]$$

For '$f_2(x)$' above integration becomes

$$x \frac{df_2(x)}{dx} - f_2(x) \Big|_{x_2} = \frac{N_2 Q_2}{3\eta}$$

$$\int dx \frac{(f_2(x))^{5/2}}{\sqrt{x}} = \frac{4}{7}\sqrt{x}(f_2(x))^{\frac{5}{2}} - \frac{5}{7Q_2}\frac{df_2(x)}{dx}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]$$

Normalization Condition:

$$\int_0^{x_{max}} dx \sqrt{x}(f_1(x))^{3/2} = \frac{N_1}{3\eta}$$

$$\int_0^{x_{max}} dx \sqrt{x}\,(f_1(x))^{\frac{3}{2}} = \frac{N_1}{3\eta}$$

$$\Rightarrow \int_0^{x_2} dx \sqrt{x}\,(f_2(x))^{\frac{3}{2}} = \frac{N_2}{3\eta}$$

From Thomas Fermi equation

$$\frac{d^2 f_2(x)}{dx^2} = Q_2 \frac{(f_2(x))^{5/2}}{\sqrt{x}}$$

$$\text{or,} \quad \int_0^{x_2} x \frac{d^2 f_2(x)}{dx^2} = Q_2 \int_0^{x_2} dx \sqrt{x}\,(f_2(x))^{\frac{3}{2}}$$

$$\text{or,} \quad \int_0^{x_2} x \frac{d^2 f_2(x)}{dx^2} = \frac{N_2 Q_2}{3\eta}$$

$$\text{or,} \quad x\frac{df_2(x)}{dx} - \int_0^{x_2} \frac{df_2(x)}{dx} dx = \frac{N_2 Q_2}{3\eta} \Rightarrow x\frac{df_2(x)}{dx} - f_2(x)\Big|_{x_2} = \frac{N_2 Q_2}{3\eta}$$

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$$\frac{4}{7}\sqrt{x_2}\,(f_2(x))^{5/2} - \frac{5}{7Q_2}\frac{df_2(x)}{dx}\cdot\frac{N_2Q_2}{3\eta} = \int_0^{x_2} dx\,\frac{(f_2(x))^{5/2}}{\sqrt{x}}$$

$$\text{(I)} = \frac{4}{7}\sqrt{x_2}\,(f_2(x))^{\frac{5}{2}} - \frac{5N_2}{21\eta}\frac{df_2(x)}{dx} = \int_0^{x_2} dx\,\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$\int_{x_2}^{x_2} dx\,\frac{(f_2(x))^{5/2}}{\sqrt{x}} = \frac{4}{7}\sqrt{x}\,(f_2(x))^{\frac{5}{2}}\Big|_{x_2}^{x_2} - \frac{5}{7Q_2}\frac{df_2(x)}{dx}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2}^{x_2}$$

$$= \frac{4}{7}\sqrt{x}\,(f_2(x))^{5/2}\Big|_{x_2} - \frac{4}{7}\sqrt{x}\,(f_2(x))^{\frac{5}{2}}\Big|_{x_2}$$

$$-\frac{5}{7Q_2}\frac{df_2(x)}{dx}\left(x\frac{df_2(x)}{dx} - f_2(x)\right)\Big|_{x_2}$$

$$+\frac{5}{7Q_2}\frac{df_2(x)}{dx}\left(x\frac{df_2(x)}{dx} - f_2(x)\right)\Big|_{x_2}$$

$$= \frac{4}{7}\sqrt{x_2}\,(f_2(x))^{\frac{5}{2}} - \frac{4}{7}\sqrt{x_2}\,(\beta_{22})^{5/2}(f_2(x))^{5/2}$$

$$-\frac{5}{7Q_2}\frac{df_2(x)}{dx}\cdot\frac{N_2Q_2}{3\eta} + \frac{5}{7Q_2}\beta_{22}^2\frac{df_2(x)}{dx}\left(x\frac{df_2(x)}{dx} - f_2(x)\right)$$

$$\int_{x_2}^{x_2} dx\,\frac{(f_2(x))^{5/2}}{\sqrt{x}}$$

From appendix A2.2

$$\text{(I)} = \frac{4}{7}\sqrt{x_2}\,(f_2(x))^{5/2} - \frac{4}{7}\sqrt{x_2}\,(\beta_{22})^{5/2}(f_2(x))^{5/2}$$

$$-\frac{5}{7Q_2}\frac{df_2(x)}{dx}\cdot\frac{N_2Q_2}{3\eta} + \frac{5}{7Q_2}\beta_{22}^2\frac{df_2(x)}{dx}\cdot\frac{N_2Q_2}{3\eta}$$

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$$\int_{x_2}^{x_1} dx \frac{(f_1(x))^{5/2}}{\sqrt{x}} = \frac{4}{7}\sqrt{x_1}(f_1(x))^{5/2} - \frac{4}{7}r_{12}^{5/2}\sqrt{x_2}(f_2(x))^{5/2}$$

$$-\frac{5}{2}\left[N_1 - N_2\left(r_{12}^{3/2} - \frac{Q_2}{Q_1}r_{12}\right)\right]\frac{df_1(x)}{dx}$$

$$+\frac{5}{2}r_{12}^2 N_2 \frac{Q_2}{Q_1}\frac{df_2(x)}{dx}$$

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